

RMO(INDIA) – 2005

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABCD be a convex quadrilateral; P, Q, R, S be the mid points of AB, BC, CD, DA respectively such that AQR and CSP are equilateral. Prove that ABCD is a rhombus. Determine its angles.
 2. If x, y are integers and 17 divides both the expression $x^2 - 2xy + y^2 - 5x + 7y$ and $x^2 - 3xy + 2y^2 + x - y$, then prove that 17 divides $xy - 12x + 15y$.
 3. If a, b, c are three real numbers such that $|a - b| \geq c$, $|b - c| \geq a$, $|c - a| \geq b$, then prove that one of a, b, c is the sum of other two.
 4. Find the number of all 5 digit numbers (in base 10) each of which contains the block 15 and is divisible by 15. (For example: 31545, 34155 are two such numbers).
 5. In triangle ABC, let D be the mid point of BC. If $m\angle ADB = 45^\circ$ and $m\angle ACD = 30^\circ$, determine $m\angle BAD$.
 6. Determine all triples (a, b, c) of positive integers such that $a \leq b \leq c$ and $a + b + c + ab + bc + ca = abc + 1$
 7. Let a, b, c are three positive real numbers such that $a + b + c = 1$. Let $\lambda = \min\{a^3 + a^2bc, b^3 + ab^2c, c^3 + abc^2\}$. Prove that the roots of the equation $x^2 + x + 4\lambda = 0$.

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